



INTERNET BASED INTELLIGENT HOSPITAL APPOINTMENT SYSTEM

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ABSTRACT—In today's competitive service industry, the technology in service systems is used in a wide range of areas. The service companies are now providing service via internet or via other computer based systems in an increasing trend day by day. Expert systems are good examples of these applications. Today, expert systems are used in various fields such as design, planning, imaging, diagnosis, etc. For practical use, the expert systems are also used through internet. One of the most important service system institutions is the hospital. Increasing the service quality level in hospitals, internet based appointment systems are used in several hospitals in Turkey. There are also several internet based expert system applications today contributing the improvement of service levels in several industries. In this study, an internet based expert system is created that is used in outpatient department/polyclinic direction. The system architecture, algorithm and the role of such an expert system are discussed in this paper. With the help of internet based expert appointment system at hospitals, the queues in the hospitals decline, the number of consultations decreases, the patients and doctors save time and finally the customer satisfaction and quality level increases.

Key Words: Internet based expert systems; Hospital appointment systems; Outpatient department/polyclinic direction

1. INTRODUCTION

In today's world, the service and manufacturing institutions are utilizing technology in an increasing trend to grow competitive. Service technologies have radically reordered the power relationships, competitive environments and opportunities in most industries – whether in services or manufacturing [1].

In the process of quality improvement, the companies have started to use a variety of techniques. Especially in recent years, with developing computer technology, artificial intelligence techniques are used in various applications [2]. One of the artificial intelligence techniques is expert systems. The evolution of expert systems as a discipline commenced in 1956 with Shannon's, Minsky's and McCarthy's studies. Expert systems (ES) are a branch of applied artificial intelligence (AI), and were developed by the AI community in the mid-1960s. The basic idea behind ES is simply that expertise, which is the vast body of task-specific knowledge, is transferred from a human to a computer. This knowledge is then stored in the computer and users call upon the computer for specific advice as needed. The computer can make inferences and arrive at a specific conclusion. Then like a human consultant, it gives advices and explains, if necessary, the logic behind the advice [3].

Expert systems have been used to solve a wide range of problems in research fields such as medicine, mathematics, engineering, geology, computer science, business, law, defense, and education. Within each field, expert systems are used for solving problems of different types. They are used for diagnosis (of a system fault, disease or error), design (of computer systems) and interpretation of data. [4].

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Table I. Some Application Areas of Expert Systems.

Function	Purpose	Application Area
Diagnosis	<i>Determination of system disorders</i>	<i>Electronics</i>
Diagnosis	<i>Diagnosis of diseases</i>	<i>Medicine</i>
Interpretation	<i>Defining the status of the sensor transmitter</i>	<i>Voice recognition, image analysis</i>
Estimation	<i>Implication of results</i>	<i>Weather forecast</i>
Design	<i>Object designing with limited sources</i>	<i>Circuit design</i>
Planning	<i>Design process</i>	<i>Automatic programming and military planning</i>
Imaging	<i>Comparison of observations for planning sensitivities</i>	<i>The regulation of nuclear power plants and cost management</i>

The first expert system was developed by Feigenbaum in 1967 named DENDRAL. This system was composed of Heuristic Dendral and Meta Dendral and was programmed with LISP [5]. After then expert systems have been used in a variety of applications from teaching to scheduling strategy, from system development to DNA histogram interpretation, etc.

Expert systems are the computer systems which can model the decision process of an expert or a number of experts [6]. Expert systems are fast growing and developing information technology [7]. Today, expert systems are used in a variety of areas, some of which are listed in Table I below:

Other application areas of expert systems are given in Bo and Monaco's [8,9] studies in detail. As seen, expert systems are used in a variety of areas as a technological improvement.

There are a lot of problems faced in service systems. Some problems in the health sector are the waste of people's time in waiting and repeating revisits, because of lack of appointment system [10]. Hospital systems as important service institutions maintaining critical service operations on human life must utilize technology for easing the activities. Hospitals serve several customers namely patients. The patient satisfaction is positively correlated with well treatment of the patients at hospitals. The patients mostly complain about the long waiting hours, disorderly time scheduling of outpatients departments, consultations, etc. Internet based appointment systems are good examples of solutions to these problems because, accurate, up-to-date information is vital to maintain the quality of healthcare [11]. Medical processes require a high level of expertise and sharing. Distribution of this medical knowledge through internet is a beneficial approach. To a great extent, this avoids unnecessary duplication of work, which may lead to tremendous social and economic savings [12]. In this study, an internet based medical expert system for a hospital appointment system is created that enables those social and economic savings.

The originality of this study comes from the application of expert systems on a web application for directing patients for the right department. This solves many problems faced in university hospitals, such as decreasing long waiting times and queuing time for patients, providing knowledge for directing patients for the right department, decreasing time spent for patient processes, decreasing work load for front-desk personnel and enabling doctors planning their time efficiently. The paper is organized as follows: In Section 2, internet based expert systems and applications in medical systems are discussed. Section 3 discusses hospital appointment systems in Turkey. In Section 4 the developed system is discussed. Section 5 discusses concluding remarks and future studies.

2. INTERNET BASED EXPERT SYSTEMS AND APPLICATIONS IN MEDICAL SYSTEMS

Expert systems are used in several fields of application such as interpretation, estimation, diagnosis, planning, design, imaging, etc. Some expert systems applications in medicine are: MYCIN [13,14] that provides a list of possible culprit bacteria ranked from high to low based on the probability of each

diagnosis; PUFF [15] an expert system for interpretation of pulmonary function data; CASNET [16], which is an expert system for consultation in the diagnosis and therapy of glaucoma; INTERNIST [17] and VM [18] broad-based computer-assisted diagnostic tool for educational experiments; ONCOCIN [19] that is designed to assist physicians in the treatment of cancer patients; PIP [20] an hierarchical based expert system for the synthesis of chemical process flow sheets; HODGKINS [21] a computerized program was used for making prognosis for each patient; HEARMED [22] psychological consultation; IRIS [23] glomcom diagnosis; and GUIDON [24], which is an adaption of MYCIN, that offers the student a tutor for medical diagnosis, etc. These expert systems used in medicine are all used for diagnostic purposes of diseases and medical decision making. These expert system applications are not internet based systems.

Today internet based expert system applications are more popular because the Internet provides a more effective base for expert systems delivery: (i) The Internet is widely accessible; (ii) Web-browsers provide a common multimedia interface; (iii) Several Internet-compatible tools for expert system development are available; (iv) Internet-based applications are inherently portable and (v) Emerging protocols support cooperation among expert system [25].

Internet based intelligent systems have been used as an information technology in a variety of areas. In the literature there are several applications of internet based expert systems. Some applications are on choosing an academic major [26], management of gallstones, renal, ureteric and bladder calculi based on ultrasound images [27], as a tool for power transmission design [28], expert-level fault diagnosis service for remote devices [29], machining operation planning in Internet environment [30], intelligent design system for injection molds [31], strategic marketing planning [32], food dryer selection [33] and groupware design [34].

More and more people are using computer-based systems in a variety of different ways for health information [35]. Some recent internet based intelligent system applications in medicine are on fish disease diagnosis [36], pig disease diagnosis [37], and breast cancer [38]. Like these systems, in the literature we see that expert systems in medicine are generally used for diagnostic purposes. Besides diagnostic purposes, the internet based systems should also be used for supporting other medical services, like in the study of Bellika et al. on an oncological nurse assistant [39]. Intelligent system application developed in this study is based on an internet based hospital appointment system as a decision support mechanism.

A web-based appointment system can facilitate the patients to make an appointment. However, the medical knowledge of the patients is limited. They cannot always make a correct decision about which department they should visit [40].

In development of medical expert systems the rule of human experts is crucial. The teams of such experts are developing expert systems to consider the changes in medicine [41]. In this study, the expert system is created with the help of doctors and professors in a research hospital. In the next section of the study the hospital appointment systems in Turkey and advantages and disadvantages of these systems are discussed.

3. THE HOSPITAL APPOINTMENT SYSTEMS IN TURKEY

When we consider a hospital as a system, the inputs are expertise, personnel (doctors and other staff), investment in financial terms, patients; the processes are medical examination, diagnosis and treatment. The outputs are healed people and treatment plans for some patients [42]. The management of this system is not so easy, because the time a doctor can spend for his/her patient is too limited in state hospitals. For example, in Turkey, the ratio of number of patients over number of doctors exceeds 2000. The doctors have limited time for patients. Thus, time saving becomes a very important improvement. Therefore in Turkey, there are several hospitals using appointment systems for facilitating the scheduling of patient visits. The patients can arrange appointments via telephone, via internet and in some hospitals they arrange their appointment when they arrive at hospitals with the help of information desks. Making appointments via telephone has several drawbacks. The user waits for 4–5 minutes for making the appointment, and then

Table II. Advantages and Disadvantages of Appointment Booking Types.

Appointment Type	Advantages	Disadvantages
Via telephone	<i>-Information desk conflict is avoided</i>	<i>Arduous to make up because long lists of outpatient department/polyclinic not easy to determine on phone Limited time for booking appointments Not suitable for hearing-impaired people</i>
Via internet	<i>No waiting to book an appointment Appointments can be made after hours Information desk conflict is avoided Easy Access Ease of use May also serve the patients with access to pertinent office information Patients' details can be checked as the appointment is made Suitable for hearing impaired people</i>	<i>Not easy for use for older people</i>
Information desks	<i>Unlimited time for booking appointments</i>	<i>Long waiting times in the hospital because of unscheduled appointments</i>

when she/he goes to the hospital, the user cannot be serviced at the scheduled time of appointment, because of disorderly schedules. Making appointments with the help of information desks is very time consuming. Arranging appointments via internet is the fastest and easiest way of this process. Although, it also has some possible drawbacks, the advantages of internet booking systems exceed the disadvantages. The advantages and disadvantages of the appointment booking types are given in [Table II](#).

When we compare the advantages and disadvantages of appointment booking types in hospitals, it is clearly seen that making appointments via internet is more advantageous compared to other methods of booking appointments. The appointment process without the help of experts is also time consuming, because of choosing wrong polyclinic and wrong operations. This causes unnecessary consultations in the hospitals. The users without the help of some experts may schedule an appointment with a wrong outpatient department/polyclinic. Directing the patients to the right outpatient department/polyclinics requires expertise. The internet based expert systems are good platforms for these kinds of problems.

The basic difference of our study from other appointment systems is that: In classical internet based appointment systems there is not any intelligent application. In the classical system: The users use the system by means of a web site. By following the standard procedure, the users fill the web page forms by entering name, surname, identity number and other required information. Then he/she selects date and doctor of his/her appointment. The classical process ends here. However, in our system there are intelligent

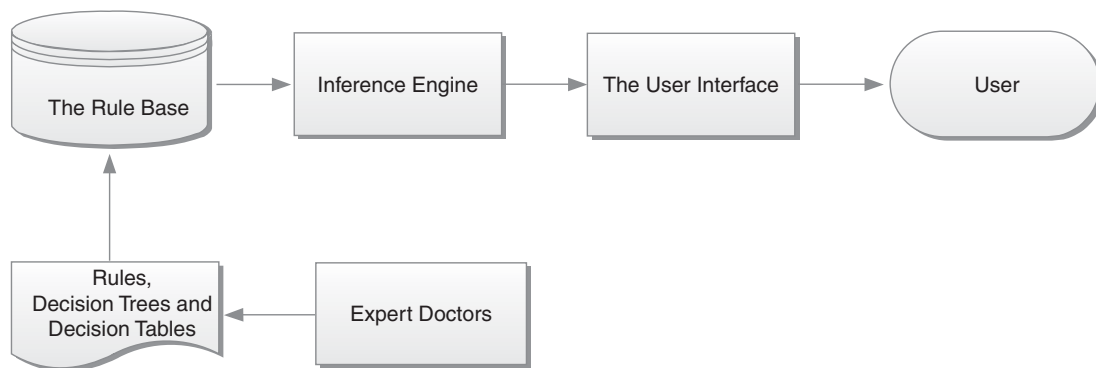


Figure 1. The Expert System Architecture.

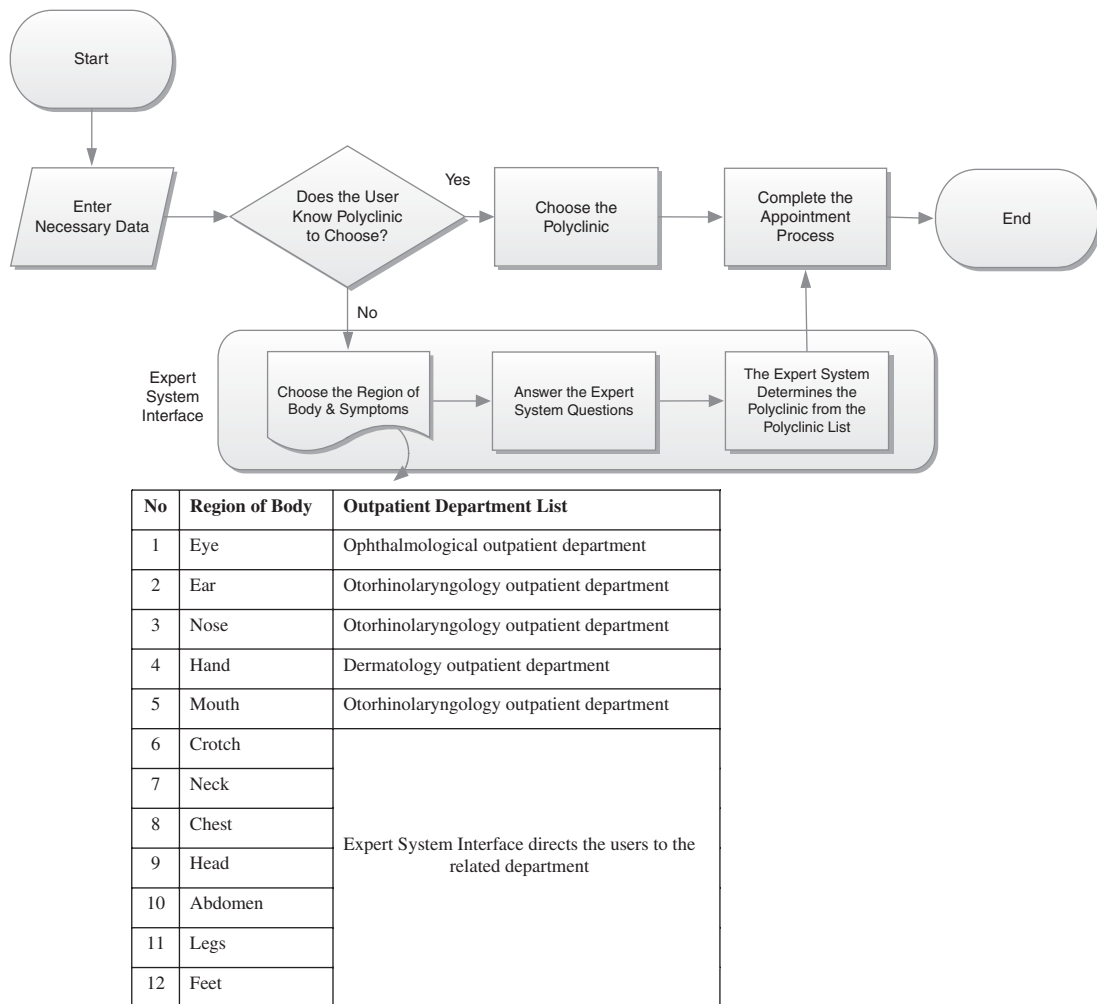


Figure 2. The Basic Algorithm of the Online Expert Appointment System.

applications for both patients and doctors. For patients: The system helps the patient in selecting which department to visit (This is a very important advantage for especially university hospitals such as decreasing long waiting times and queuing time for patients, providing knowledge for directing patients for the right department and decreasing time spent for patient processes, decreasing work load for front-desk personnel). For doctors: Enabling doctors planning their time efficiently is another advantage. The system enables doctors to schedule their appointments.

In the next section, the design, development and implementation of the internet based intelligent hospital appointment system is discussed.

4. INTERNET BASED EXPERT HOSPITAL APPOINTMENT SYSTEM

Different from the use of expert systems in medicine for diagnostic purposes, the system that is designed in this study does not direct the user for the determination of diagnosing diseases. The system directs the users who

want to make an appointment via internet. The users who don't know which outpatient department/polyclinic to choose before visiting the hospital is directed by the expert system embedded in the online appointment system.

The rules and decision trees of the expert system are created with the help of expert doctors and professors. The rule base of the system is created with these rules and decision trees. The inference engine interprets the rules and transfers them to the user interface. The user interface is a web site. The expert system architecture is given in Figure 1.

The expert system application discussed in this paper is like a consulting system. When the user enters the web site for booking an appointment, the user sees a human body figure. The user then clicks on the related region of the body from the human body figure on the web page. For regions 1, 2, 3, 4 and 5 given in a table in Figure 2, the patient is directed to the related outpatient department/polyclinics. For other regions of the body, the expert system interface works and some questions are asked to the user and the system directs him/her to the related outpatient department/polyclinic. For instance, a patient who chooses the head part of the human body figure (meaning having complaints about head region) answers the questions and the expert system directs him/her to the related outpatient department/polyclinic.

The basic algorithm of the internet based appointment system with an expert system interface is given in Figure 2.

The rule base of the whole system is created with the help of expert doctors and professors in a research hospital. The decision tree and table of the outpatient department/polyclinic selection for the patients having problems with abdomen region is given in Figure 3 above as an example. The rule base of the

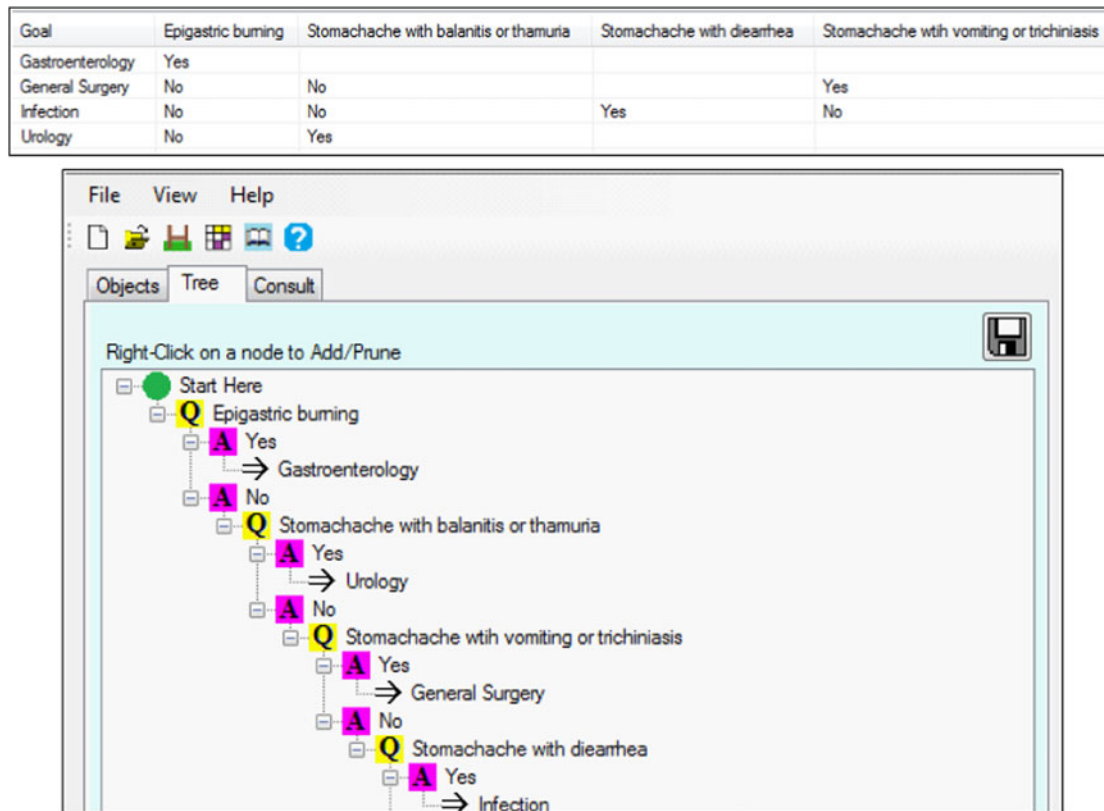


Figure 3. Decision Table and Decision Tree of the Polyclinic Direction System as an Example for Abdomen Region.

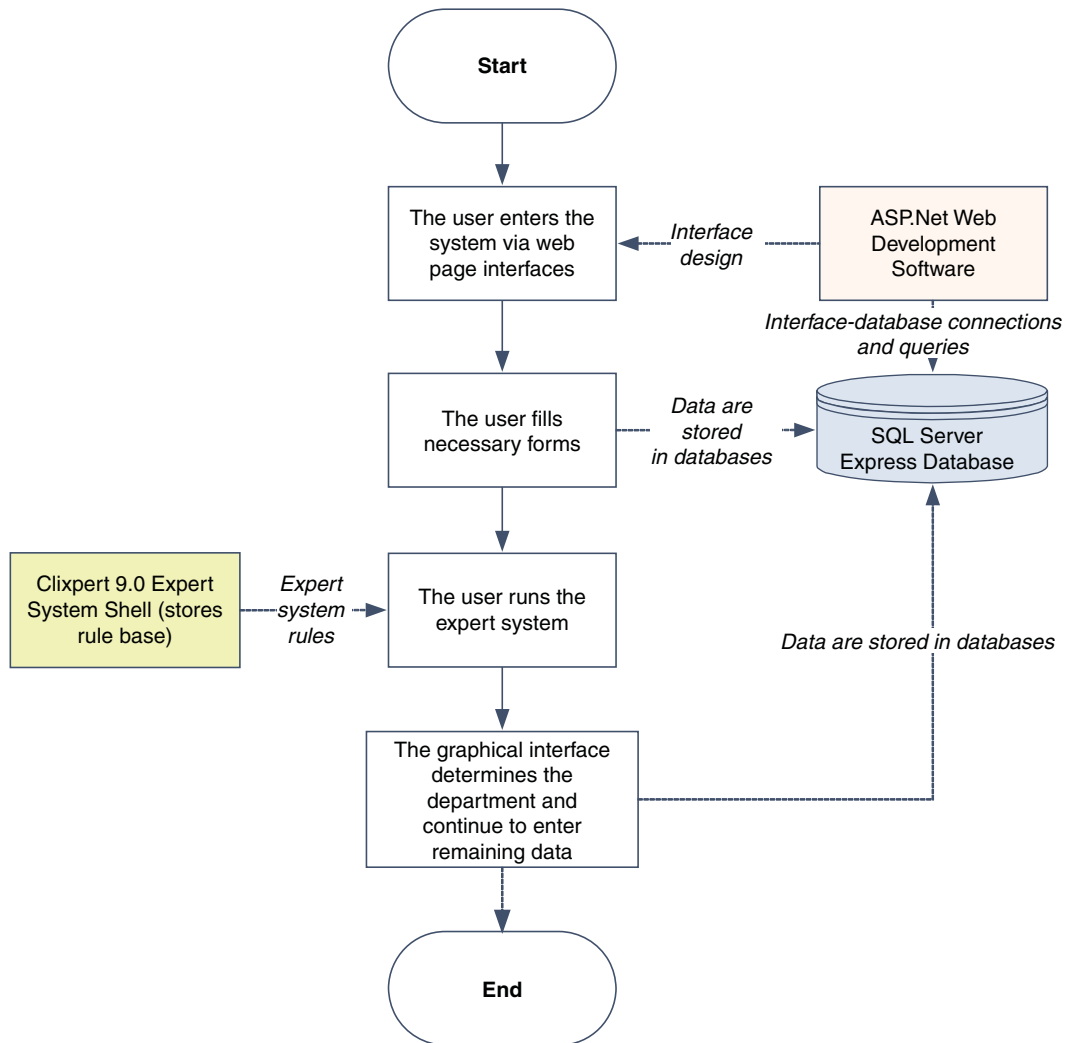


Figure 4. System Interactions.

system is supported with Clixpert 9.0 expert system shell [43]. And the user interface is developed as an ASP.Net application (<http://www.asp.net/>) as a website and the queries is constructed with SQL Server Express Database (SQL Server Express Database Design Software). The whole system interactions are displayed in Figure 4.

First, the users (patients and doctors) enter the system using an interface developed by ASP.Net software. Data entered to the system by patients or doctors are stored in SQL server database, which is connected to ASP.Net platform. After users enter necessary log-in information they use the expert system, which is embedded in appointment software. The expert system rules are created with Clixpert 9.0 expert system shell. Here rules created by Clixpert 9.0 are converted to a graphical user interface (human body figure shown in Figure 6). Graphical user interface is embedded in ASP.Net platform.

In Figure 5, the main page of the user interface is given. There are 4 links on the main page: The first link is for booking the appointment and completing the appointment process. The users can book the day

and time of appointment here. The second link is about membership to the web site. The users must sign up with their national ID number and a user name. The third link is the expert system interface, that enables the patients choose the right polyclinic for the appointment process. When the user clicks on this link, as seen in Figure 6, the user firstly chooses the region of the body that he/she has complaints. Then the user chooses the symptoms from the menu and answers the expert system's questions. Finally, the expert system directs the user to the right polyclinic and the fourth link is about help videos of the web site's usage.

As mentioned, appointment booking via internet is advantageous. The intelligent appointment system given in this study is more advantageous both for patients and doctors, because it provides economic and time savings. With the help of this system, doctors and patients will use their time more efficiently.

Use of the internet by healthcare consumers is not limited to public websites and information searching, but can be used to complement treatment in the form of online self-help support programs, and provide patients with access to telemedicine and other care and monitoring systems [44]. Our system can also be used as a self-help support program because people can learn about their health problems and which symptoms cause which illnesses by using the intelligent system.

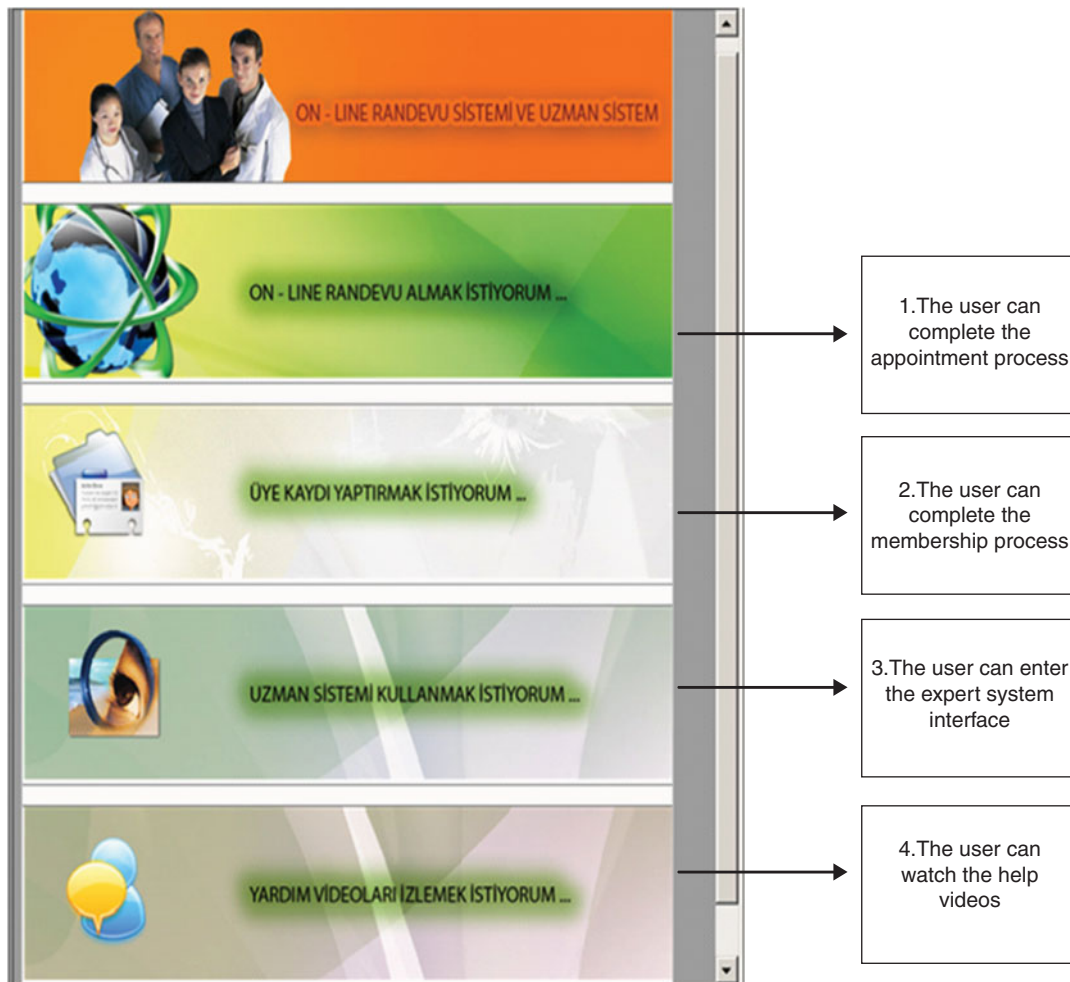


Figure 5. The Internet Based Expert Hospital Appointment System.

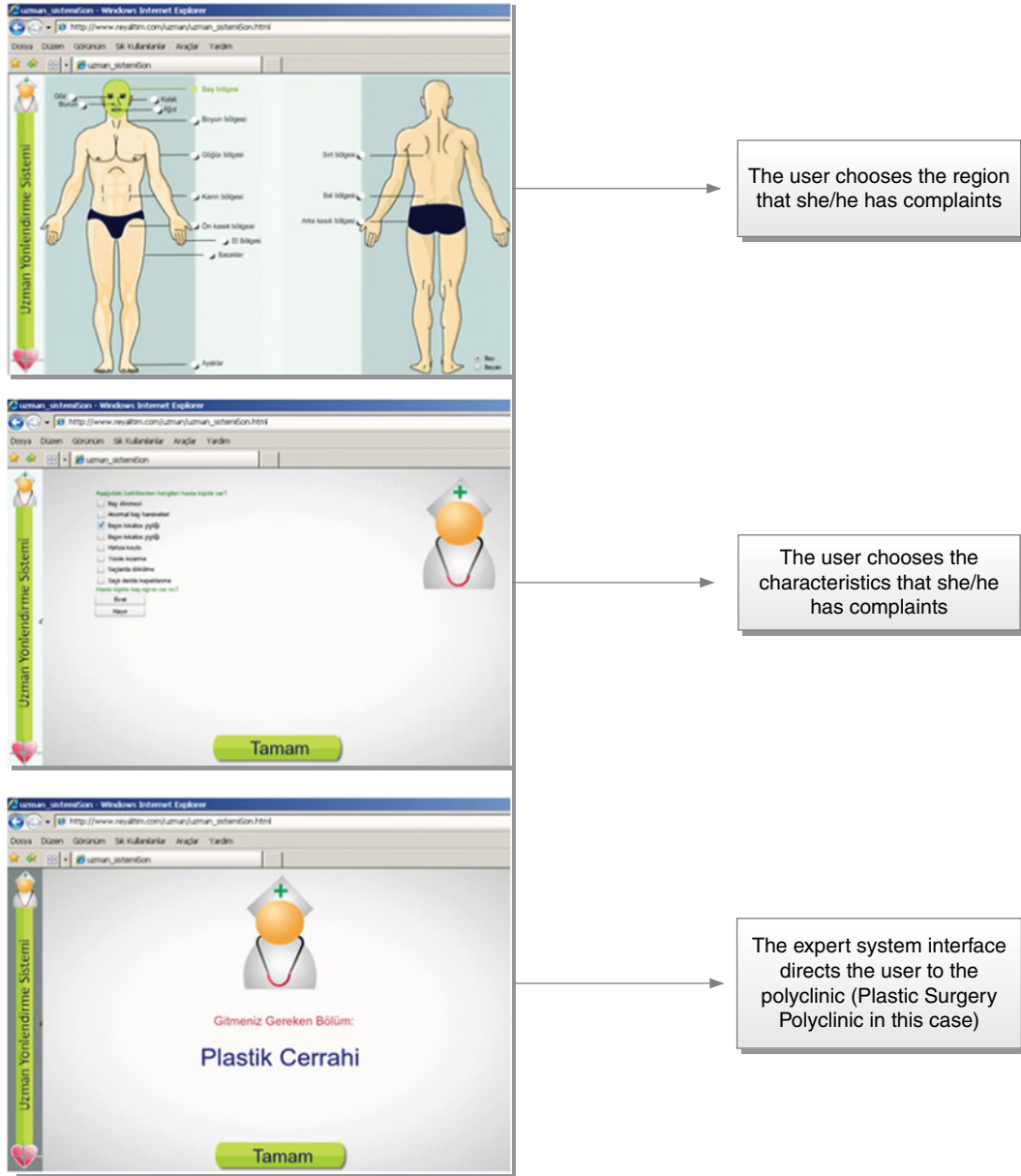


Figure 6. Polyclinic Direction with the Help of Expert System (Graphical User Interface).

5. CONCLUSION

The use of technology in both manufacturing and service sectors is increasing day by day. One of the computer based systems, expert systems, as a branch of artificial intelligence are used in a variety of areas from manufacturing applications to service industry applications.

A hospital as an important service institution, because of conducting critical activities on human life uses appointment systems for scheduling appointments in the hospital. Thanks to the appointment systems, the queuing problem can be solved, time savings can be accomplished and the service quality levels can be increased in the hospitals.

In this study, an internet based expert hospital appointment system is created and benefits of this system are discussed. With the help of this appointment system, the users are able to make appointments with the help of the expert system interface and the users are directed to choose the right outpatient department/polyclinics. The online expert appointment system provides a lot benefits such as shorter waiting times in hospitals, decrease in the number of consultations and saving both patients' and doctors' time. In addition, service rates can be standardized in the outpatient department/polyclinics.

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